BME 261 Final Report

Development of a Ruggedized Myoelectric Prosthetic Hand for High-Load Operational Tasks in Heavy-Duty Environments

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Executive Summary

This project aimed to develop a practical myoelectric arm for Edward, a firefighter who experienced a right transradial amputation following a fire-related accident and wishes to return to fieldwork. The selected Cybathlon-inspired tasks included gripping water bottles, placing them into a crate, and carrying the crate. These tasks represent common gripping and lifting challenges in firefighting. The main design requirements included reliable gripping, simple user control, stability under load, passive load retention, and operation using only one motor.

The final design consists of a thumb and four fingers combined into two interconnected sections. These sections can bend around an object to create a more adaptable grip. The mechanical subsystem uses sprockets, chains, gears, rotating shafts, and springs to coordinate the movement of the finger sections. A dog clutch controlled by a servomotor lock releases the mechanism, allowing the hand to maintain its grip without requiring continuous motor operation.

The signal-processing and electrical subsystems convert the user's forearm muscle activity into a control signal for the servomotor. The circuit consists of an instrumentation amplifier, a band-pass filter, a buffer, a non-inverting amplifier, a peak detector, and a comparator. Together, these stages amplify and process the EMG signal and determine whether the muscle activation exceeds a preset threshold. The firmware then moves the servomotor between two preset positions to engage or disengage the dog clutch.

Following several rounds of design, prototyping, and testing, the team completed an integrated prototype that successfully demonstrated the core gripping, locking, and EMG-based control mechanisms. However, the prototype was not fully successful as a final solution for the intended firefighting application because it did not meet the required durability, heat resistance, and reliability. Future development should focus on using stronger and more heat-resistant materials, improving mechanical tolerances and connections, reducing the control delay, and conducting additional testing with different objects and loading conditions.

I. Introduction

Edward is a professional firefighter who experienced a right transradial amputation approximately 14 months ago and hopes to return to active duty (Figure 1.1). Although he is new to prosthetic control systems, he is familiar with advanced rescue equipment and motivated to learn. However, many existing upper-limb prostheses are designed mainly for appearance or daily activities and may not meet the demands of firefighting.

Myoelectric arms are electrical power devices. They use electrodes over the residual forearm muscles to detect signals produced during muscle contraction [1]. These signals are processed to control motor-driven functions such as gripping and releasing [1]. This allows the user to operate the arm using their remaining muscles, but sweat, electrode movement, electrical noise, and sudden movements may reduce control reliability.

Firefighters work in environments involving heat, water, humidity, soot, dust, and impact, and may suddenly shift from rest to intense physical effort [2]. Edward therefore needs an arm that responds quickly, tolerates forceful movements, and securely holds heavy or uneven objects [3]. An unreliable grip could cause dropped equipment or lead to overuse of his unaffected arm, contributing to shoulder fatigue and lower-back pain. His main needs include heat and impact resistant construction, rapid attachment, reliable gripping, passive load retention, and a wrist that can lock [3]. The arm must also be simple to control during emergencies and capable of supporting the variety of gripping, lifting, and carrying tasks involved in firefighting.

Figure 1.1. An AI-generated image of firefighter Edward, our persona, in the firefighting field, ready to rescue residents trapped in the building. (Credit: Nano Banana 2)

To represent the variety of lifting and carrying tasks associated with firefighting, our team benchmarked the project against the Cybathlon "Carry Bottles" task, which requires picking up weighted bottles, placing them into a crate, and transporting the load under time pressure [4]. The primary goal was to develop a functional myoelectric hand capable of passively opening and conforming around bottle and crate handle geometries, securely locking around the objects, and releasing them under active user control. To maximize energy efficiency and safety, the hand was designed to retain loads mechanically without requiring continuous motor torque once locked. The scope was governed by course constraints, most notably the restriction to a single position servomotor, which prevented independent finger driving. Secondary requirements identified for the persona, such as rapid socket attachment, wrist rotation, thermal shielding, and temperature feedback, were excluded from this prototype phase and are recommended for future development. Production methods and materials were selected primarily to demonstrate core mechanical and electrical functions within the course timeline.

The mechanical subsystem includes a thumb, a palm, and four fingers combined into top and base phalanges blocks. Because only one motor could be used, the servomotor was reserved for controlling the locking mechanism rather than directly driving the fingers. When the hand is pressed downward against the water bottle, friction and contact forces cause the finger sections to open and conform to its shape. For the crate task, the hand is pushed against the rim of the crate, and the resulting reaction force passively forces the fingers to stretch and conform to the geometry of the handle. Chains, sprockets, gears, and rotating shafts coordinate the movement of the finger sections. When forearm flexion is detected, the position servomotor actively engages the dog clutch. This prevents the transmission shafts from rotating and locks the fingers in their current position. Once engaged, the dog clutch mechanically retains the grip without requiring continuous motor torque. When the servomotor disengages the clutch, the shafts can rotate again, allowing springs to return the finger blocks toward their initial positions. The plywood structure and casing also act as physical stops that restrict the maximum rotation angles.

The electrical subsystem detects the user's forearm muscle activity through an EMG input. Because the original EMG signal is small and contains noise, it passes through an instrumentation amplifier, a band-pass filter, a non-inverting amplifier, a peak detector, and a comparator. These stages amplify and condition the signal before producing an output that can be interpreted by the microcontroller.

The firmware reads the processed control signal and moves the servomotor between two preset positions. At one position, the servomotor engages the dog clutch to lock the mechanical system. At the other position, it disengages the dog clutch, allowing the springs to return the finger sections to their initial configuration. In this way, the three subsystems convert muscle activation into a mechanical locking or releasing action while using only one motor.

II. Mechanical Subsystem

The solution's mechanical design focused on 2 main objectives. First, the fingers must move continuously from a flat, open position, to a closed circular position (Figure 2.1). This addresses the prosthetic constraint of needing to conform around a circular object, such as a water bottle or handle for carrying. The second objective is that the fingers must lock into place, at a desired position. This is necessary since the prosthetic must be able to carry an object. Without a locking mechanism, the weight of an object will manually push the fingers back, unintentionally releasing it. This mechanism is controlled by a position servo integrated into the electrical subsystem, giving the user control of object release.

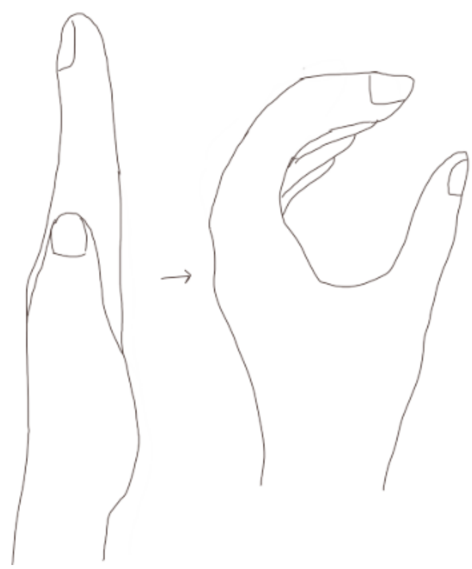


Figure 2.1. Visualization of hand movement from open to closed position.

To address both objectives, an initial prototype was designed (Figure 2.2). In the first design, finger movement is motor driven. It incorporates a ratchet that is rotated by a motor. During rotation, strings threaded through the four fingers are pulled or released, opening and closing the fingers. To achieve the locking constraint, a pawl is used to prevent ratchet rotation and consequently finger movement in the outward direction. This prevents the fingers from being pushed backwards while carrying tasks. To unlock the mechanism and allow the prosthetic to flatten back out, and eccentric cam is rotated to engage the pawl and lift it off the ratchet.

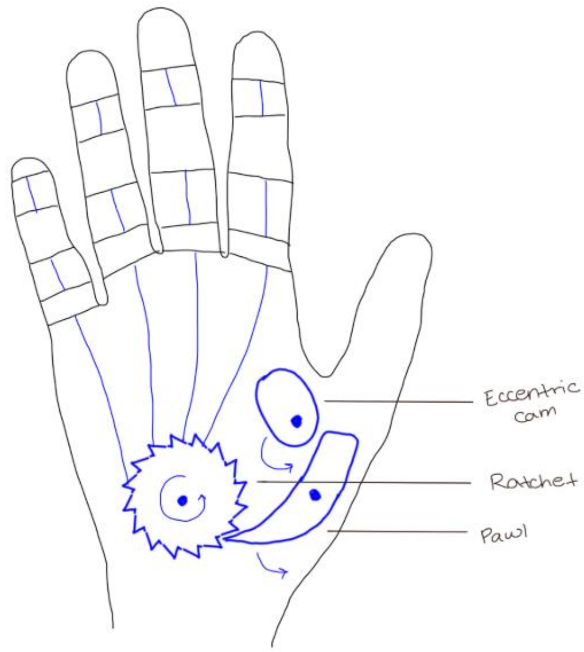


Figure 2.2. Initial sketch of prosthetic design. Motor driven ratchet pulls on strings to open and close fingers. Pawl utilized for locking mechanism and released through rotation of eccentric cam.

Upon receiving feedback through the gallery walk, extensive changes were made to this design. Since all four fingers move simultaneously, a question was raised about the benefit of having separate fingers. Upon analyzing this feedback, future iterations adapted to having connected fingers to allow for greater strength and simpler mechanics due to the individual and larger finger component. Other feedback surrounded the locking mechanism, and it was apparent that a single motor could not rotate both the ratchet and the eccentric cam in two different ways as initially intended. The ratchet needs continuous rotation, while the eccentric cam should only be rotated in and out of place to lock or unlock finger position. So, this whole mechanism was updated.

After analyzing all the feedback and revisiting the constraints, a second iteration was designed (Figure 2.3). Due to the singular motor limitation, the motor is used solely for locking mechanisms. The pawl, screwed directly into the position servo, can be moved between two set positions. When the pawl engages with the teeth of the worm wheel, the fingers are locked into place. When the pawl moves away from the worm wheel, the fingers are free to move freely. As a result, finger movement is mechanical, rather than motor driven. The overall design uses spring loaded finger that holds the fingers in a closed position at rest (Figure 2.4). When wrapped around a circular object, like a water bottle or handle, the fingers are manually pushed back, stretching the spring. Upon user activation through a forearm muscle contraction, the fingers will be locked in place at their current position, preventing the fingers from being flattened further. This allows the object to be carried. The other major change made to this iteration is in how the fingers move. Rather than being controlled by pulling strings, sprockets and bike chains are used to connect the palm, base phalange, and top phalange. They are connected such that when the base phalange is pushed back, the top phalange also flattens. This allows the prosthetic to move continuously between a semi-closed fist, optimal for fitting around a circular object, and a flat palm, allowing for objects of multiple sizes to be carried. Since the tasks specifically centers around circular objects, the inside of the hand contains a curved surface to provide a better fit.

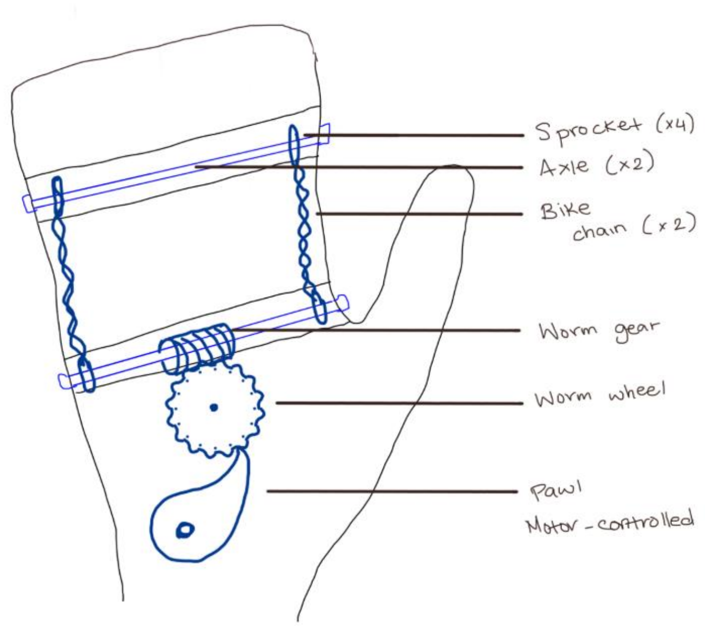


Figure 2.3. Sketch of the second iteration.

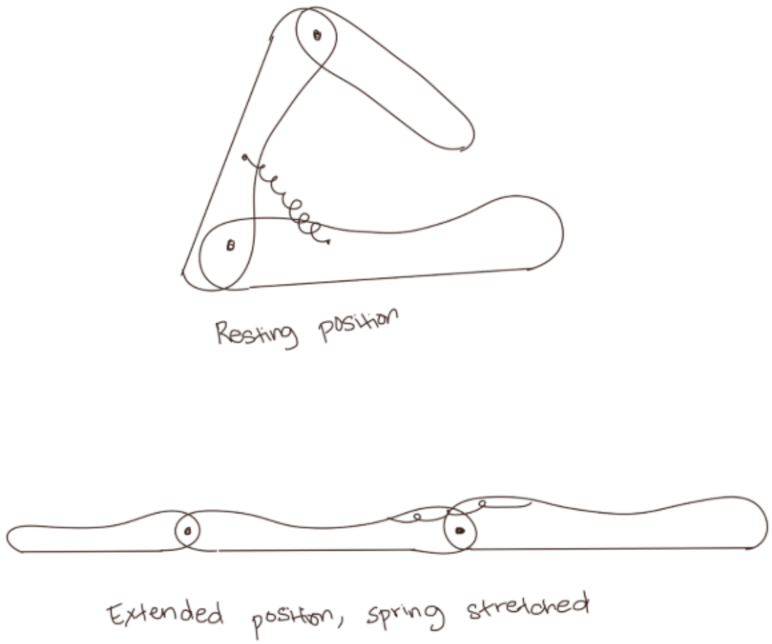


Figure 2.4. Spring loaded fingers. Held in a semi-closed fist at rest. The springs are manually stressed to flatten the hand.

To test the efficacy of this design, a low fidelity prototype was created (Figure 2.5). Cardboard was used to create the casing of the prosthetic, while the internal components were 3D printed for rapid production. Upon inspection of this physical model, various limitations were discovered. Due to the gear ratio between the worm gear and worm wheel, the full movement of the prosthetic between the open and closed position caused very little rotation of the worm wheel. As a result, the pawl was ineffective at stopping movement, since the mechanism relied on stopping the worm wheel in a specific position. To address this, bevel gears were used since their gear ratio can be altered. Another issue was observed in the finger movement. As the base phalange is moved inwards towards the closed position, the intention was for the top phalange to also move inwards. However, due to the chain connected between the sprockets for each phalange, the top phalange moved in the opposite direction of the bottom one. To fix this issue, additional gears were added in future iterations to change the direction of rotation.

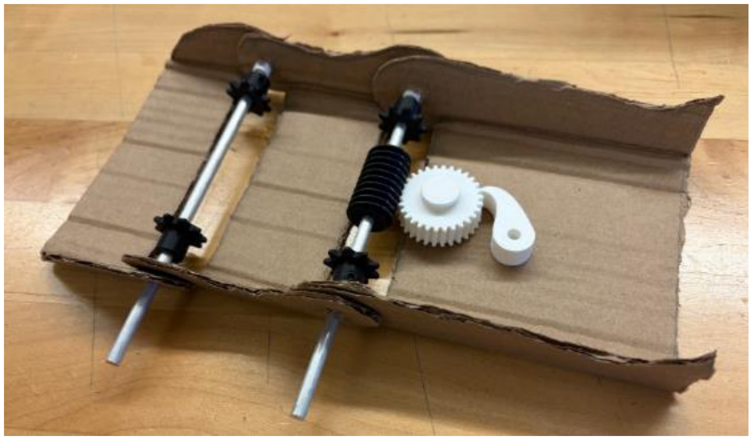


Figure 2.5. Physical prototype of second iteration. Uses pawl and worm wheel to lock phalanges in place.

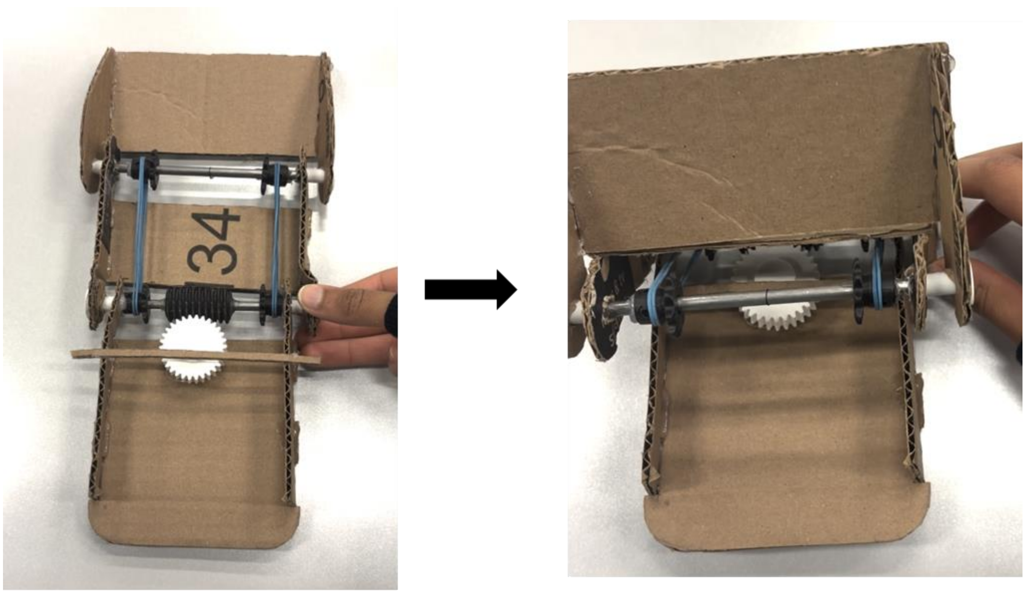


Figure 2.6. Error in finger movement. As the bottom phalange is moved into the closed position, the top phalange incorrectly flips upwards.

After troubleshooting the first physical model, a third iteration was prototyped (Figure 2.7). While this prototype fixed the identified problems, other issues arose. When the fingers were spring loaded, the torque caused slipping in two key areas, the interface between the bevel gears and the pawl, and gears connected to the top and base phalange. This prevented the fingers from locking reliably. To address the gear slippage between the two phalanges, the gear module was increased (Figure 2.8). The thicker tooth profile increased contact area and stability. To improve the reliability of a locking mechanism, the final iteration used a dog clutch. This preserves the concept of using the position servo to move a component between locked and unlocked positions. However, it greatly increases the surface area, creating a stronger grip on the base axle to stop movement.

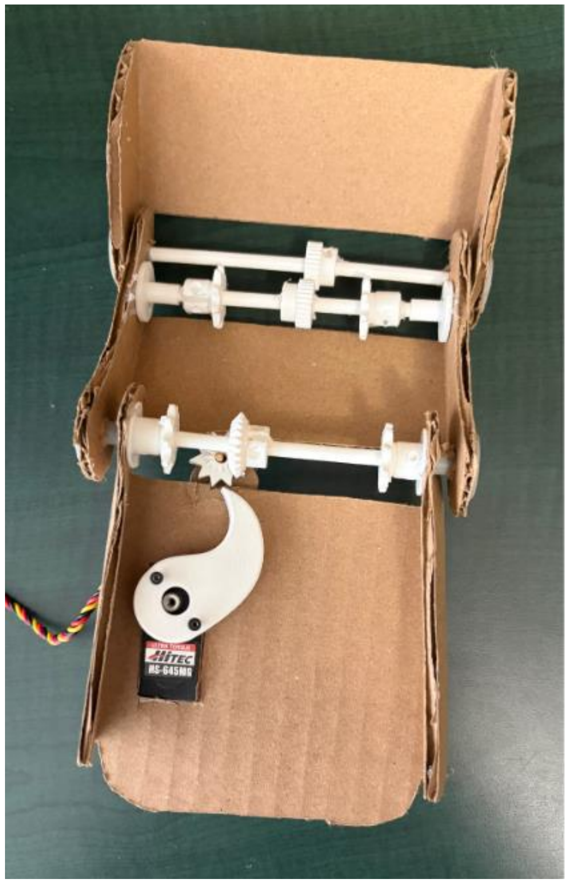


Figure 2.7. Physical prototype of the third iteration. Uses bevel gears to address the gear ratio problem. Uses additional gears to change the direction of rotation for the top phalange.

Figure 2.8. Increased gear module to prevent slipping between the top and base phalange.

Through analyzing the limitations revealed by the first 2 low fidelity prototypes, a final prototype was constructed (Figure 2.9). Laser cut plywood with box joint connections were used to increase the mechanical strength of the prosthetic casing. Additionally, aluminum axles replaced the steel axles used in the first physical prototype to decrease the overall weight while maintaining sufficient strength. A casing for the electrical subsystem was added to allow for better integration. Overall, this design addressed the defined constraints through its ability to control the movement of the base and top phalange between the flat and closed positions and lock the fingers in a desired position (Figure 2.10).

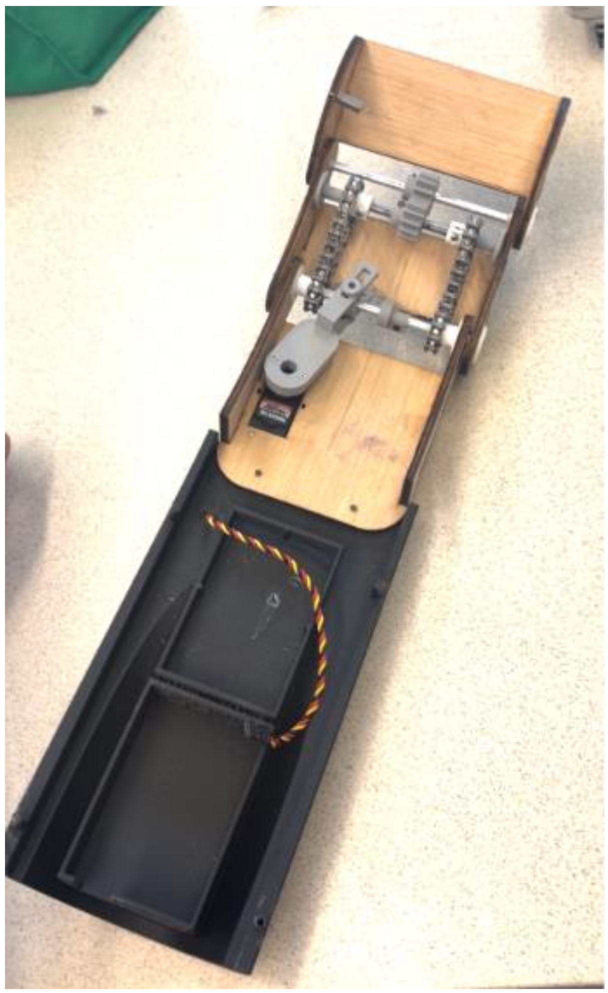


Figure 2.9. Final prosthetic prototype. Includes improved gear module, dog clutch locking mechanism, plywood base, and electrical casing.

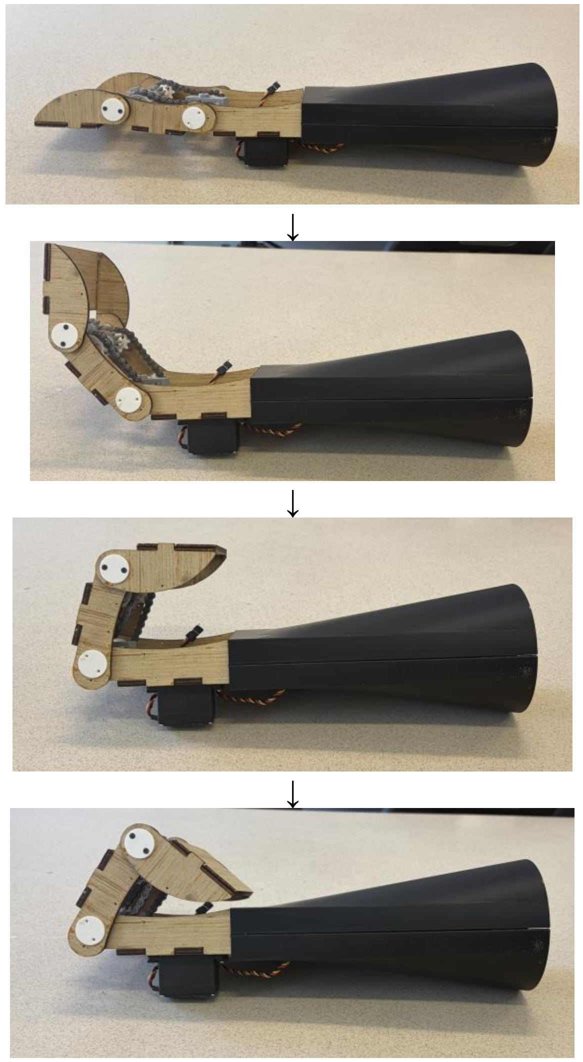


Figure 2.10. Controlled movement of the base and top phalange between the closed and flat positions. When the base phalange is moved, internal gears move the top phalange accordingly.

III. Signal Processing and Firmware Subsystem

Surface EMG signals captured during muscle contraction are inherently noisy, highly stochastic, and prone to rapid high-frequency fluctuations. Raw analog comparator outputs can chatter near the decision threshold due to motion artifacts or minor muscle twitches, which can cause erratic servo motor jitter, false triggering, or mechanical binding, making user control impossible without secondary signal smoothing [5].

The goal of the firmware subsystem is to convert analog signals into smooth, predictable servo motor actuation. The firmware must reliably distinguish intentional, sustained muscle contractions from transient noise spikes while enforcing safe motor travel bounds. To achieve this, the control logic leverages contextual timing cues tailored to the specific analog performance characteristics and non-idealities of the electrical subsystem.

To ensure optimal performance and use safety, the firmware subsystem was developed around four operating constraints First, to prevent accidental contraction from high-frequency noise, or muscle twitches, the algorithm requires an intentional muscle contraction to persist continuously for an activity window of 500 milliseconds (~0.5 seconds). While this introduces a half-second processing delay, it prioritizes false-trigger immunity and trigger confidence over hyper-responsiveness. Next, to ensure the motor stops after a contraction, the system must enforce a mandatory quiet period of 400 milliseconds where no pin toggling occurs before considering a muscle burst finished, preventing rapid, erratic re-triggering during a single contraction. Moreover, Servo sweep angles must be strictly constrained in software between 135 degrees (hand open) and 150 degrees (hand closed) to prevent physical joint binding, gear stripping, or the extreme 2.5A stall current draws observed near 170 degrees. Finally, the logic must implement a state lockout to ensure that a single sustained muscle flex results in exactly one state toggle rather than continuous rapid cycling.

The embedded firmware is written in C++ and runs on an Arduino microcontroller. Rather than tracking raw analog amplitude or relying on hardware interrupts, the software evaluates muscle activity using a non-blocking, time-domain state machine. The control algorithm continuously polls the digital output of the hardware comparator stage to monitor signal activity. When a user flexes a muscle, high-frequency voltage fluctuations pass through the comparator, causing the digital input pin on the Arduino to rapidly toggle between zero volts and five volts. Rather than treating a single rising voltage edge as an immediate trigger command, which would leave the hand vulnerable to static shocks or brief muscle twitches, the system treats state changes as candidate activity. Upon detecting any state transition, the program starts an internal timer to measure the duration of the contraction. The execution loop then continuously evaluates two simultaneous conditions. The first condition is the active contraction handling, if high-frequency toggling persists continuously for at least 500 milliseconds (activityWindow), the algorithm recognizes a deliberate, sustained muscle contraction. It flips the internal state of the hand, commands the servo motor to move to its target angle, and immediately sets an execution lock to prevent continuous re-triggering during the remainder of the flex. The second condition handles the rest identification, where if the signal stops toggling and remains static for longer than 400 milliseconds (quietTimeout), the algorithm determines that the muscle has returned to a relaxed state. It clears the execution lock and resets the state machine so the hand is ready to receive the next intentional contraction. This dual-timer architecture effectively acts as a digital state-machine filter, absorbing threshold noise from the analog front-end and translating binary muscle activity into smooth, predictable hand control. The flow chart outlining the firmware logic is showcased in figure 3.1

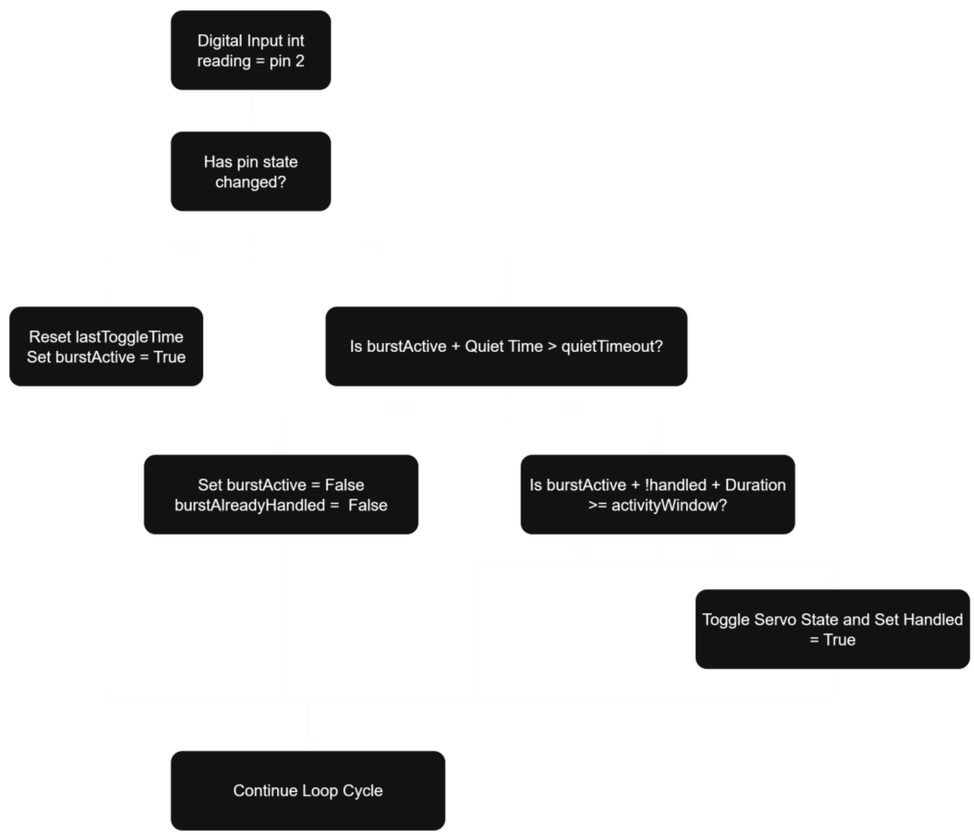


Figure 3.1: Time-domain burst-detection algorithm flowchart. The state machine monitors input pin edge transitions to track candidate muscle bursts, qualifying intentional contractions after a 500 ms continuous activityWindow and resetting the execution lock once a 400 ms quietTimeout rest phase is observed.

As shown in figure 3.2, the execution loop constantly polls inputPin. By comparing the current reading against lastReading, the code detects toggling across both signal edges. If activity begins while burstActive is false, the program records burstStartTime and initializes the candidate burst tracking.

```
    // detect toggling (any change counts as "activity", not just rising edges)
    if (reading != lastReading)
    {
      lastToggleTime = now;

      if (!burstActive)
      {
        burstActive = true;
        burstStartTime = now;
        burstAlreadyHandled = false;
      }
    }
```

Figure 3.2: Edge State Detection and Burst Initialization

While a burst is active, the program continuously verifies whether the muscle contraction is sustained long enough to qualify as a deliberate input. As detailed in figure 3.3, once the duration (now - burstStartTime) meets or exceeds activityWindow (500 ms) and burstAlreadyHandled is false, the firmware toggles servoIsFlexed, writes the new target angle to the motor, and sets burstAlreadyHandled = true.

```
    // if we're in an active burst that's lasted long enough, and haven't handled it yet, trigger the toggle
    if (burstActive && !burstAlreadyHandled && (now - burstStartTime >= activityWindow))
    {
      servoIsFlexed = !servoIsFlexed;

      if (servoIsFlexed)
      {
        myServo.write(openAngle); //135 degrees
      }
      else
      {
        myServo.write(closedAngle); //150 degrees
      }

      burstAlreadyHandled = true; // don't re-trigger again until this burst ends and a new one starts
    }
```

Figure 3.3: Burst Duration Verification and Servo Actuation

Finally, when the user relaxes their arm, signal toggling stops. Once the time elapsed since the last edge transition (now - lastToggleTime) exceeds quietTimeout (400 ms), the algorithm clears burstActive and burstAlreadyHandled, resetting the control loop to accept the next muscle contraction. This complete firmware state machine successfully bridges the analog front-end and mechanical actuation, providing a robust, noise-immune software filter that guarantees safe, repeatable prosthetic hand control.

IV. Electrical Subsystem

Surface EMG signals originate as microvolt to low-millivolt physiological action potentials generated across skeletal muscle fibers during depolarization. When recorded non-invasively via surface skin electrodes, these signals present extreme conditioning challenges. Typical EMG amplitudes range from merely 10μV to 5mV peak-to-peak, requiring substantial low-noise amplification before they can be processed by standard microcontroller digital logic. Furthermore, these microvolt biopotentials are buried beneath significant environmental and physiological interference. Additionally, low-frequency movement artifacts caused by electrode displacement or skin stretch introduce severe baseline wander (<20Hz), while broad-band thermal noise (>500Hz) degrades the clarity of the signal. Beyond signal conditioning, powering a biomedical mechatronic system introduces severe power quality and ground stability challenges. While initial benchtop development relies on tethered laboratory DC power supplies, a truly functional upper-limb prosthetic requires autonomous, portable power. However, transitioning a sensitive analog signal chain to battery power introduces significant voltage rail constraints. Ultimately, the design must bridge delicate analog signal extraction with heavy transient power demands inside a single, portable architecture.

The physical signal conditioning pipeline processes microvolt-level muscle signals through a cascaded, seven-stage single-supply analog architecture. As illustrated in the block diagram in Figure 5.1, biopotentials move sequentially through differential instrumentation, active bandpass filtering, isolation buffering, secondary voltage gain, peak envelope detection, and threshold comparator stages before interfacing with the microcontroller.

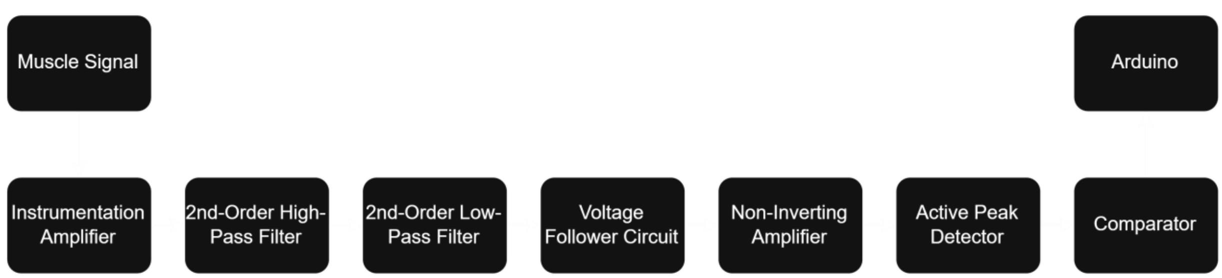


Figure 5.1: Block diagram of the seven-stage analog signal conditioning pipeline, depicting signal flow from surface electrodes through instrumentation amplification, active bandpass filtering, buffer isolation, non-inverting gain, peak envelope detection, and comparator logic thresholding.

The physical hardware chain begins with an AD623 instrumentation amplifier connected directly to surface skin electrodes. Configured with a precision 1k ohm gain resistor, this initial stage delivers a differential gain of roughly 101, maximizing the common-mode rejection ratio to suppress ambient 60 Hz mains interference before subsequent amplification. The reference pin of the instrumentation amplifier is tied directly to a buffered virtual ground midpoint, centering the output signal swing symmetrically between the power rails. Filtering is accomplished using a pair of active two-pole Sallen-Key filter networks designed to isolate the clinical electromyography frequency spectrum. First, the signal enters a Sallen-Key high-pass filter configured with two 33nF capacitors, a 1M ohm resistor, and a 10k ohm resistor to establish a corner frequency of approximately 48 Hz at unity gain, aggressively stripping away low-frequency motion artifacts and electrode drift. The signal then passes into a Sallen-Key low-pass filter utilizing two 20k ohm resistors, a 12nF capacitor, and a 33nF capacitor to set a high-frequency cutoff of roughly 400 Hz at unity gain, eliminating out-of-band thermal noise. Because both filter stages are designed with heavily overdamped quality factors, the transition around the cutoff frequencies is gradual rather than a sharp, abrupt corner. The theoretical design response alongside the measured frequency response featuring this overdamped transition shape is depicted in the Bode plot in Figure 5.2.

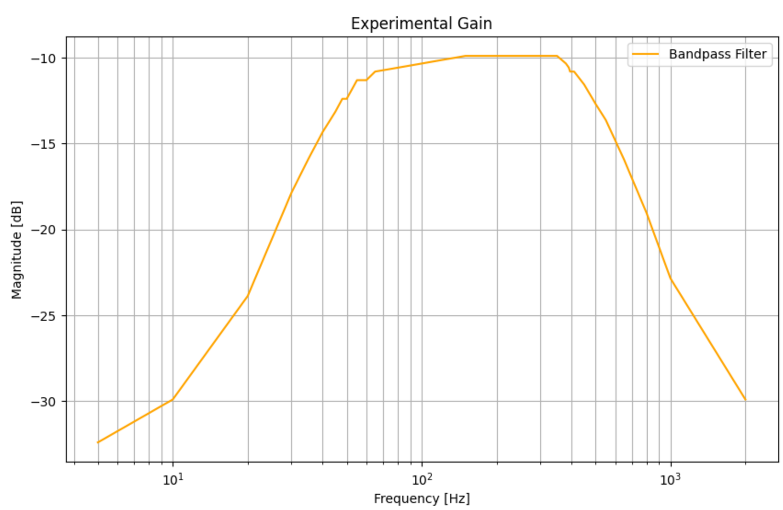
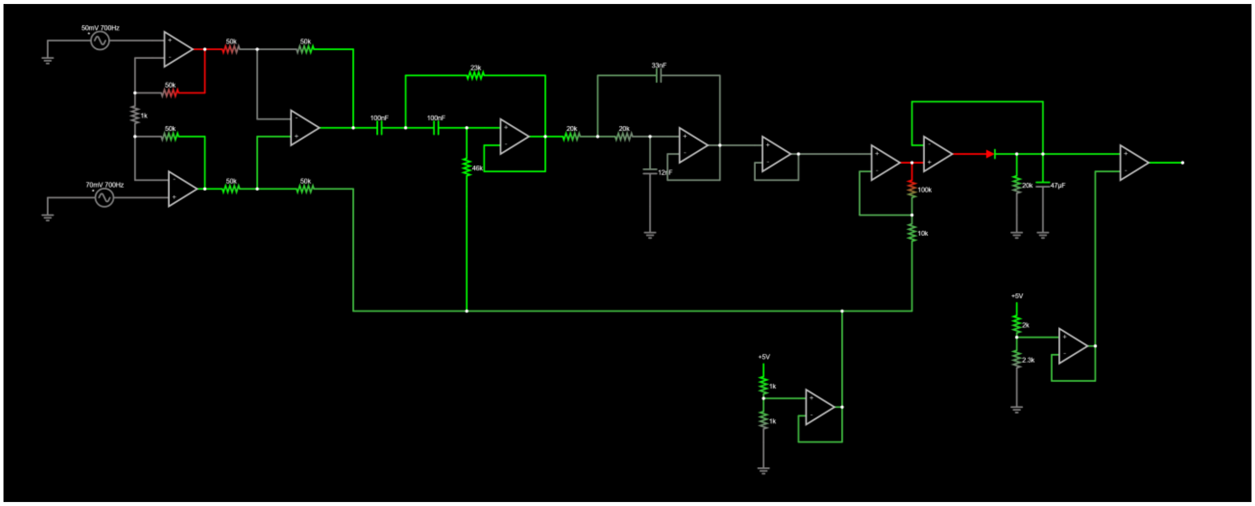
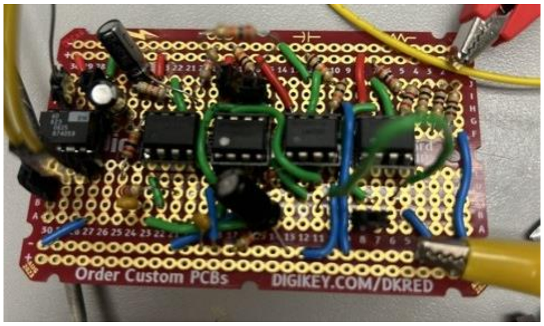


Figure 5.2: Experimental frequency response Bode plot of the active bandpass filter cascade. The plot shows a flat passband gain, with gradual transition roll-offs near the corner frequencies due to heavily overdamped high-pass and low-pass filter responses.

Following the active filter stages, an operational amplifier voltage follower serves as an isolation buffer. This buffer prevents input impedance loading from downstream gain stages from pulling down signal amplitude or shifting the filter corner frequencies. Next, a non-inverting operational amplifier stage configured with a 10k ohm input resistor and a 100k ohm feedback resistor provides a secondary voltage gain of 11. Across the first five stages, total AC analog system gain reaches approximately 1,111 times, converting millivolt-level muscle action potentials into multi-volt AC signals centered around the virtual ground reference. To prepare the high-frequency AC muscle bursts for threshold logic evaluation, the signal passes into a peak envelope detector stage. This active diode-capacitor circuit rectifies and smooths the oscillating AC waveform into a continuous DC envelope voltage that tracks the amplitude profile of the muscle contraction. Finally, a Schmitt trigger comparator evaluates this smooth DC envelope against the secondary voltage divider reference to generate a clean, chatter-free binary logic signal for the digital input of the Arduino microcontroller. The complete hardware schematic detailing all component values, op-amp connections, and stage layouts is provided in Figure 5.3.





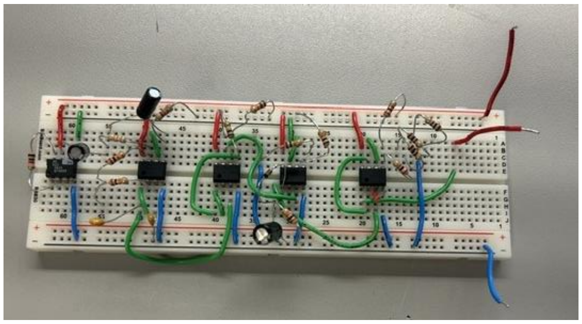


Figure 5.3: Complete system schematic diagram, breadboard and perfboard showing component values for the AD623 instrumentation amplifier, Sallen-Key filter networks, buffer, non-inverting amplifier, peak detector, Schmitt trigger comparator, and virtual ground circuit.

To operate the entire analog front-end from a single positive power source, a virtual ground buffer creates a stable DC midpoint reference set to half of the supply voltage. Two 10k ohm metal-film resistors divide the positive supply rail in half, which is then buffered by a low-impedance op-amp follower. This ratio design guarantees that whether the circuit is powered by a regulated 5V laboratory supply or a dynamic 6V battery pack, the virtual ground reference automatically scales to exactly 50 percent of the supply voltage. Additionally, a secondary voltage divider utilizing 2k Ω and 2.3k Ω resistors was implemented to establish a separate, dedicated reference voltage tailored specifically for the Schmitt trigger comparator thresholding. Consequently, no manual threshold recalibration or gain adjustments are required when switching power sources.

Although initial benchtop development relied on tethered laboratory power supplies, functional portability testing prompted the introduction of a dual 4xAA battery pack connected in parallel to provide roughly 6V with doubled current capacity. Under laboratory conditions, a regulated DC power supply provides superior voltage stability and near-zero rail ripple, making it ideal for precision analog measurements. For mobile operation, the parallel battery architecture was designed to provide sufficient current capacity to support motor actuation without dropping the logic supply below the microcontroller reset threshold, though full-load verification under simultaneous motor stall conditions remains an unverified limitation of the physical prototype. Ultimately, this single-supply analog conditioning chain and dynamic virtual ground architecture reliably extract microvolt muscle biopotentials and convert them into clean control triggers, bridging delicate biopotential sensing with portable power demands.

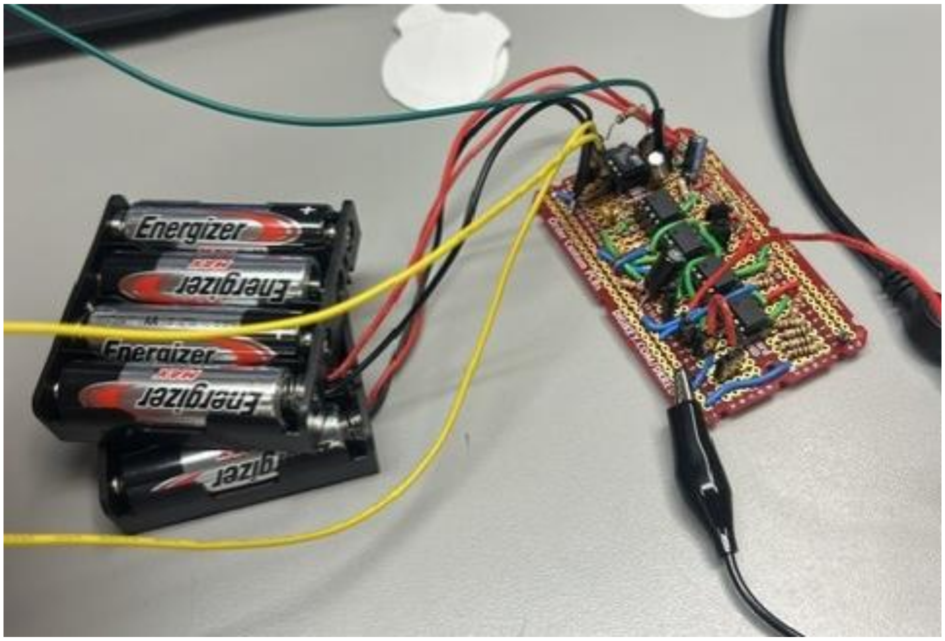


Figure 5.4: Physical parallel battery configuration featuring dual 4×AA battery packs wired in parallel (~6V) to double current capacity for portable system operation.

Ultimately, this seven-stage analog conditioning chain and dynamic virtual ground architecture reliably extract microvolt muscle biopotentials and convert them into clean control triggers, bridging delicate biopotential sensing with portable power demands.

V. Conclusion & Recommendations

In conclusion, the prototype met the main functional constraints of the Cybathlon "Carry Bottles" task. The sprocket and chain system moved the base and top phalanges, while the object load stretched the springs and opened the hand. The position motor moved the dog clutch into the locking position and held the hand in place.

The electrical subsystem amplified the muscle signals, stabilized the output, and shifted it into the 0 to 5 V range required by the firmware. The bandpass filter was designed for a frequency range of 48 to 400 Hz. The firmware moved the motor to 135 degrees during forearm extension to disengage the dog clutch and to 150 degrees during flexion to engage it. Requiring the activity condition to remain present for 500 ms also reduced false triggers caused by brief muscle twitches and electrical noise.

However, the prototype did not meet several criteria for firefighting use. The selected materials were not sufficiently heat resistant or resistant to deformation. The gears and sprockets were printed in PETG and attached to the shafts with super glue, causing the mechanism to loosen during testing. This limited its strength and long term durability.

The 500 ms activity window reduced responsiveness and made the gripping operation less intuitive. The hand soldered perfboard also required more enclosure space and used long connections that were more vulnerable to stray capacitance, human error, and electromagnetic interference. In addition, the unregulated battery supply could cause the reference voltage and signal baseline to drift as the battery discharged.

Overall, the project validated the core mechanical, electrical, and firmware concepts. However, the prototype did not fully meet the durability, heat resistance, responsiveness, and reliability requirements of the intended firefighting application.

To improve the design in the future, more research should be conducted on the materials used to better address the persona's constraints. The materials should resist deformation from heat and prevent heat transfer to the user's body due to the likelihood that the persona will need to use the prosthetic during a fire. Additionally, the material of the palm should be adapted to provide greater friction. The current, smooth, plastic palm makes it easy for objects to slide out of the prosthetic clutch. A softer material with greater friction will allow the prosthetic to better conform to objects. Currently, the strength of the locking mechanism relies on the dog clutch's ability to prevent the base axle from rotating and how well the dog clutch stays in the intended position. The plate of the clutch is secured to the axle through a nut and bolt, to tightly close the plate around the axle, and superglue. Unfortunately, if the torque applied by the object due to its weight exceeds the strength of dog clutch connection, the plate slips on the axle, and the fingers can move. Altering the connection method by drilling through the axle and inserting a pin to securely prevent the disk from rotating freely from the axle will strengthen the locking mechanism. Additionally, the springs are the only components pulling the fingers inward. So, the grip strength of the design relies entirely on the strength of the string. The locking mechanism only prevents the fingers from sliding further back. Thus, a design that can support stiffer springs without impeding the locking mechanism will improve the grip force.

To reduce the delay caused by the 500 ms activity window while maintaining resistance to false triggers, future firmware could replace binary comparator triggering with continuous Analog-to-Digital Converter (ADC) sampling. Digital filtering and a shorter moving average could then be implemented in C++ to distinguish intentional muscle contractions from noise and improve the locking response. If the mechanical system is later redesigned to actively control finger movement, muscle contraction intensity could also be mapped to servo speed and grip force.

For the electrical subsystem, the analog front-end could be transferred from a hand-soldered perfboard to a multi-layer surface-mount device (SMD) PCB. This would reduce the enclosure volume, shorten signal paths, and decrease electromagnetic interference. A dedicated Low-Dropout (LDO) regulator could also provide low-noise power to the analog front-end, while a buck-boost converter could supply the servo motor. Together, these changes would provide more stable power as the battery discharges.

Lastly, to more closely resemble a natural human hand, additional curves could be added to the casing to reduce its boxy appearance. A skin-like cover could also conceal the mechanical components and provide a smoother, more realistic surface. More natural proportions, colors, and surface textures could further improve overall appearance.

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